Low-Cost & Low-Power Multispectral PAR Sensor Node Based on TCS3448 for Precision Agriculture

Jingjie Nan , H00422554

HERIOT WATT UNIVERSITY
UK | DUBAI | MALAYSIA

Photosynthetically Active Radiation (PAR, 400–700 nm), quantified as PPFD (μmol·m⁻²·s⁻¹), is fundamental to plant photosynthesis and precision agriculture. This project developed a low-cost multispectral PAR sensor using nRF52833 and TCS3448, replacing expensive commercial sensors with a regression-calibrated, 14-channel alternative.

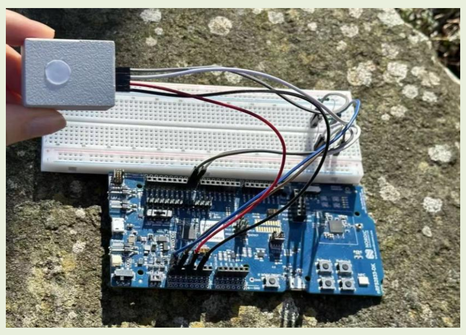
Aim & Objectives

Aim: Develop the PAR section of a compact, low-power wireless multisensor node in precision agriculture.

Objectives:

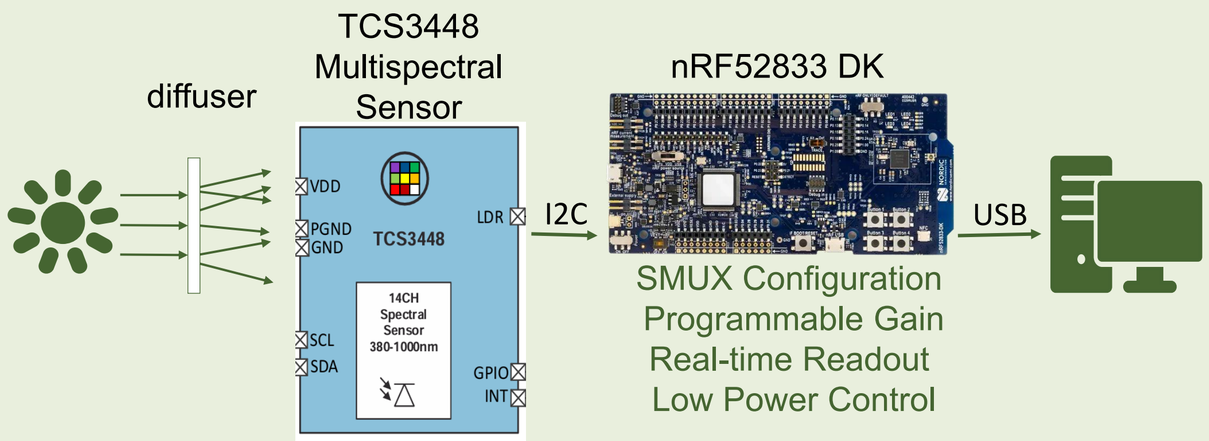
Implement efficient I²C driver and SMUX configuration for TCS3448.
Characterize a low-cost diffuser for cosine-corrected measurement.
Validate system performance and power efficiency.

Methods

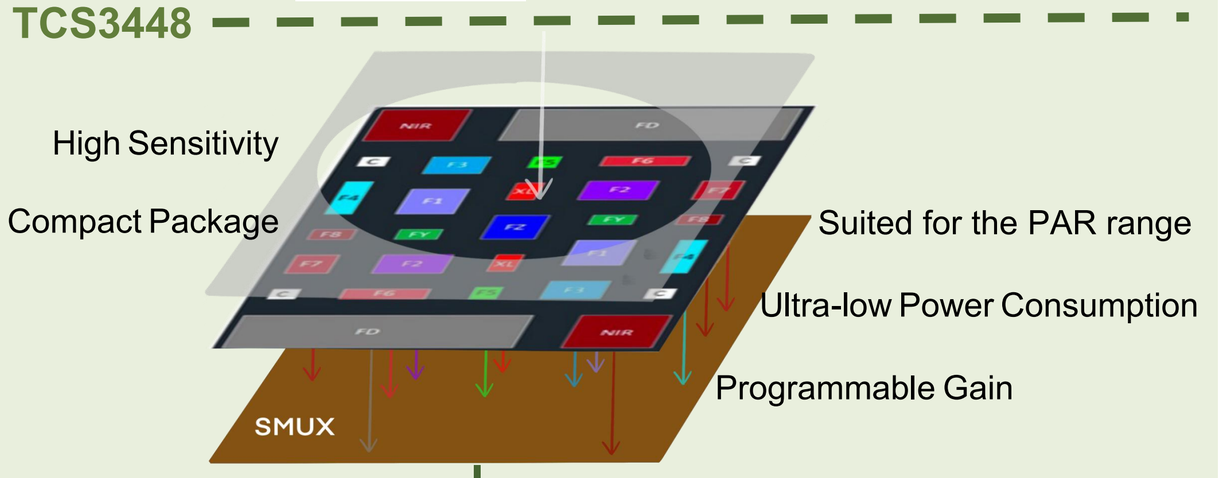


The system firmware was developed on the nRF52833 platform with a custom I²C driver. SMUX mapping, programmable gain and real-time 14-channel raw data readout have been implemented. The design focuses on low-power operation at 1.8 V using sleep mode and interrupt-driven control.

System architecture



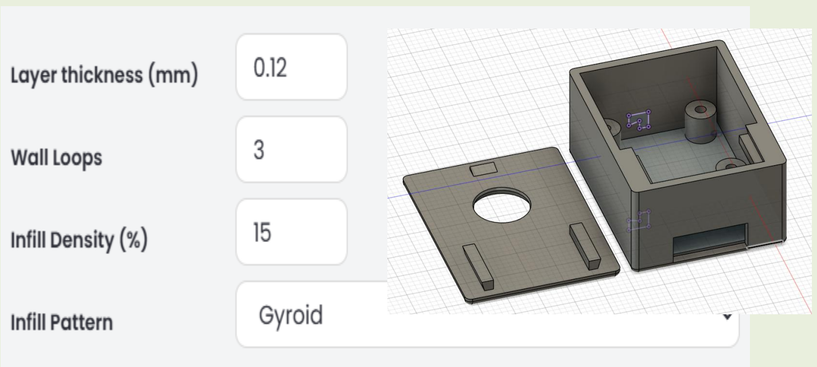
TCS3448



	ADC 1	ADC 2	ADC 3	ADC 4	ADC 5	ADC 6
Cycle 1	FZ	FY	FXL	NIR	VIS	FD
Cycle 2	F2	F3	F4	F6	VIS	FD
Cycle 3	F1	F7	F8	F5	VIS	FD

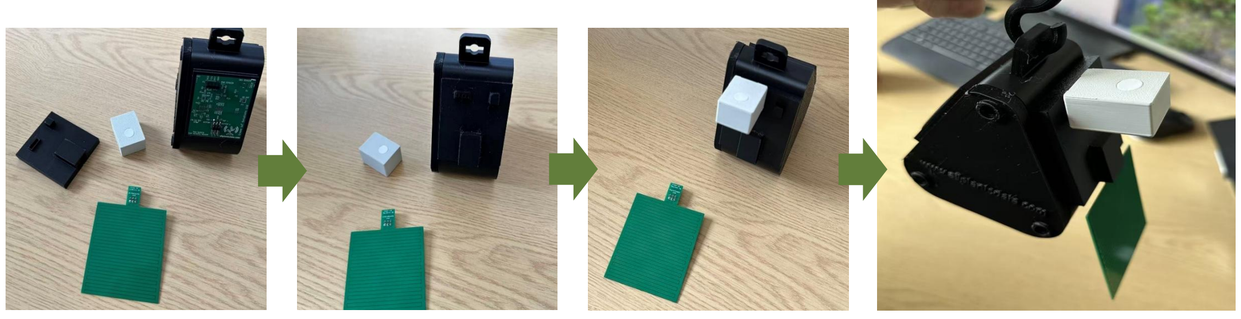
→ 18 x 16bit Data Register → **Raw Counts**

3D-printing shell



◄ Shell modeling and 3D printer parameter settings

Assembly of final product



Results

Output

Bäumker's Calibration Method:

Obtain calibration coefficients through multiple linear regression

$$PAR = b_0 + \sum_{i=1}^{n} b_i \cdot x_i$$

```
<info> app: TCS3448 Spectral Data:
<info> app: Output 0: 126
<info> app: Output 1: 252
<info> app: Output 2: 253
<info> app: Output 3: 20
<info> app: Output 4: 304
<info> app: Output 5: 7388
<info> app: Output 6: 19
<info> app: Output 7: 139
<info> app: Output 8: 153
<info> app: Output 9: 246
<info> app: Output 10: 302
<info> app: Output 11: 7340
<info> app: Output 12: 5
<info> app: Output 13: 169
<info> app: Output 14: 21
<info> app: Output 15: 66
<info> app: Output 16: 302
<info> app: Output 17: 7331
```

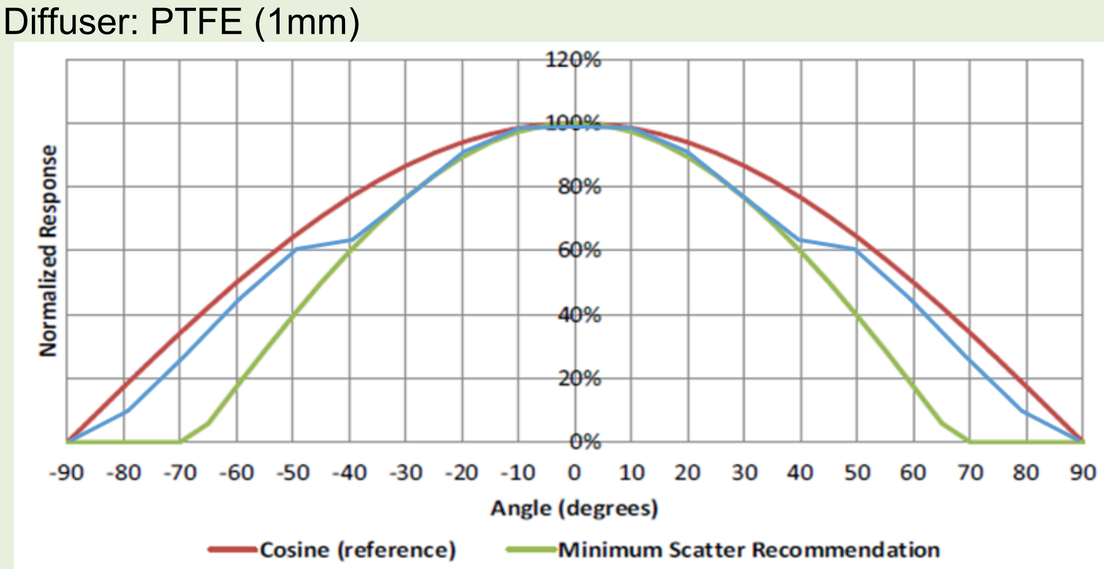
Debug Terminal

```
<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 806
<info> app: Sample: 118 (103 mV)

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 636
<info> app: Sample: 118 (103 mV)

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 372
<info> app: Sample: 120 (105 mV)

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 190
<info> app: Sample: 118 (103 mV)
```

MLR Coefficient	b0	b1	b2	b3	b4	b5	b6	b7	b8
Value	-4.02	-5.40	-0.34	4.88	-2.00	-6.83	5.34	-1.62	-5.49

Cosine Correction

Diffuser: PTFE (1mm)



▲ The cosine response curve (blue) meets the minimum recommendation

Comparation

	TCS3448 Node	Apogee SQ-521
Supply	1.8 V	5.5 to 24 V
Quiescent Current	40 μA	1.4 mA
Active Current	210 μA	1.8 mA
Cost	< £10	>£200

▲ Lower energy consumption & lower price than commercial sensors

Conclusion

A low-cost, low-power multispectral PAR sensor node based on TCS3448 has been successfully developed. The system achieves stable real-time acquisition of all 14 spectral channels via a custom I²C driver with complete SMUX configuration and programmable gain control on the nRF52833 platform. Its power consumption is already competitive with commercial alternatives; integration of a sleep algorithm is expected to reduce average current to the low-μA range. A PTFE diffuser was characterized and found to meet cosine correction requirements, enabling reliable outdoor deployment.

Reference

1. Code: daskals/nRF52833-I2C-AS7341-Low-Power-Spectral-Sensing
2. Bäumker E, Zimmermann D, Schierle S, Woias P. A Novel Approach to Obtain PAR with a Multi-Channel Spectral Microsensor, Suitable for Sensor Node Integration. Sensors. 2021; 21(10):3390. https://doi.org/10.3390/s21103390
3. TCS3448 Datasheet published by ams-OSRAM AG, TCS3448-14-Channel-multi-spectral-sensor.pdf
4. Datasheet published by Apogee Instruments, SQ-521-manual.pdf